

Final Project 2026: Introduction to Computer Vision

This project focuses on semi-supervised dataset creation and the training of a smaller model suitable for deployment on an edge device.

Overview

When starting a computer vision project, finding or creating a labeled training dataset is often the first challenge you will face. In this project, you will experience this challenge firsthand and train a model for object detection.

- You will use the [Florence-2](#) or [SAM3](#) (Segment Anything Model 3) model, along with the [Flicker image dataset](#), to create a training dataset for an **object detection** task.
- The large Florence-2 or SAM3 model will be used to create training data for a smaller model using one of the following frameworks:
 - [MMDetection](#)
 - [Ultralytics](#) (excluding YOLO WORLD and YOLOE)
 - [Torchvision / Native PyTorch](#)

Goals

- Experience real-world challenges when training a model for object detection.
- Create training data without human annotation (semi-supervised).
- Select a model that can run on edge devices (for example: NVIDIA Jetson or Raspberry Pi).
- Train an object detection model on a custom dataset.

Deliverables

- Presentation (PPTX).
- Video clip from each team member presenting the slide deck.
- Interview (optional): The course staff reserves the right to interview each student regarding their deliverables.
- The trained model weights.
- Source code for training the model.
- Source code for creating the training dataset.
- Source code for inference on a single image and on multiple images in a folder.

Tasks

1. Creating the Training Dataset

- Use the Florence-2 or SAM3 model (in inference mode) and the Flickr dataset to create annotations for the smaller model's training dataset. You may use the provided Colab notebooks ([Florence-2](#) / [SAM3](#)) as a baseline.
- You have degrees of freedom regarding the minimum and maximum object sizes, as well as the minimum and maximum image sizes.
- Dataset requirements:
 - The training dataset should include bounding boxes for the following classes:
 - **Person** (woman, man, boy, girl, child...)
 - **Vehicles** (cars, bike, motorcycle)
 - The dataset must contain at least 500 training images and 100 validation images per class.
 - Write how you could use TTA (Test Time Augmentations) to check the correctness of the annotations (**not required to implement**).
 - Write how ensembles can be used to check the correctness of the annotations of the training images (**not required to implement**).
 - Explain your decisions regarding the degrees of freedom (object/image sizes).

2. Training the Model

- Choose one of the following models (or model families). Explain why you chose it (including the specific model from the family) and why you dismissed the other architectures:
 - Faster R CNN
 - RetinaNet
 - YOLO family
 - EfficientDet
 - MobileNetV4 (**Note:** *This is advanced work that includes the integration of the MobileNetV4 backbone with your selection of a detection head*).
 - Another model of your choice (requires prior approval from the course staff).
- **Choose one of the following PyTorch-based frameworks for training:**
 - **Ultralytics:** Includes all YOLO models and SAM3. Recommended for out-of-the-box YOLO models and easy export to Edge formats.
 - **Torchvision / Native PyTorch:** Includes Faster R-CNN and RetinaNet. Recommended for students who want to build the detector head over a custom backbone from scratch.
 - **MMDetection:** Includes Faster R-CNN, RetinaNet, and EfficientDet. Recommended for modularity and easy integration of MobileNetV4 via [timm](#).
- Train the model on your custom dataset. Explain its performance, providing training graphs and relevant metrics.
- Apply appropriate augmentations and re-train (or continue training) your model.
- Explain your thought process and the insights gained during training.

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Presenting Results

- Prepare a slide deck suitable for a 5-minute presentation (if working individually) or a 7-minute presentation (if working in pairs).
- Suggested presentation structure (the number of slides is a recommendation; adjust as needed):
 - **Explain the problem:** (1 slide)
 - **Dataset creation:** (1-3 slides)
 - Explain your decisions regarding degrees of freedom.
 - Explain the challenges you faced and how you overcame them.
 - Explain how you could have used TTA and ensembles to enhance dataset quality.
 - **Model training:** (1-3 slides)
 - Explain why you chose your specific architecture.
 - Show the results of the raw model (trained without augmentations) versus the results after applying augmentations.
 - Show 1-3 representative exemplary images.
 - Show the Mean Average Precision (mAP) of the models.
 - Support your performance claims with training graphs.
 - **Future work:** (1 slide)
 - What would you have done additionally if you had more time?
 - **Reflection:** (1 slide)
 - What are your conclusions from this project?
 - What did you learn?
 - What would you have done differently?

Video Submission

- Record a video clip (up to 5 minutes) of yourself presenting the slide deck.
Note: While the project and class presentations can be done in pairs, the recorded video must be made individually by each student.